

ABERDEEN 2040

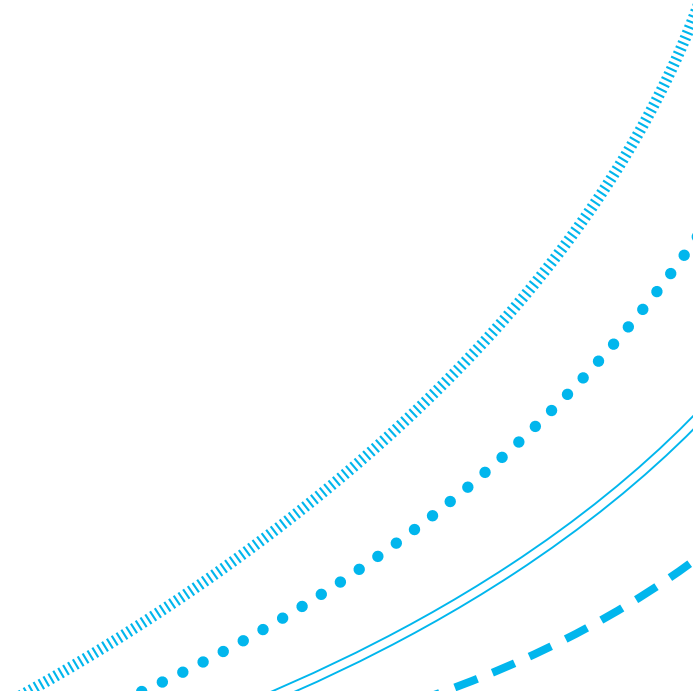
Relationships In Data

Data Mining & Visualisation
Lecture 5



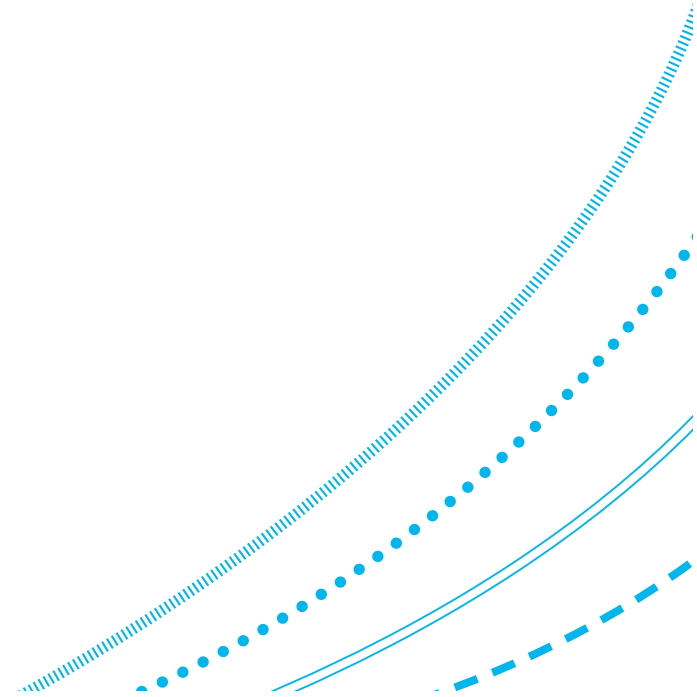
Recap...

- Categorical vs. Quantitative data
- Levels of measurement



Today...

- Relationships in data
- Pearson Correlation Coefficient

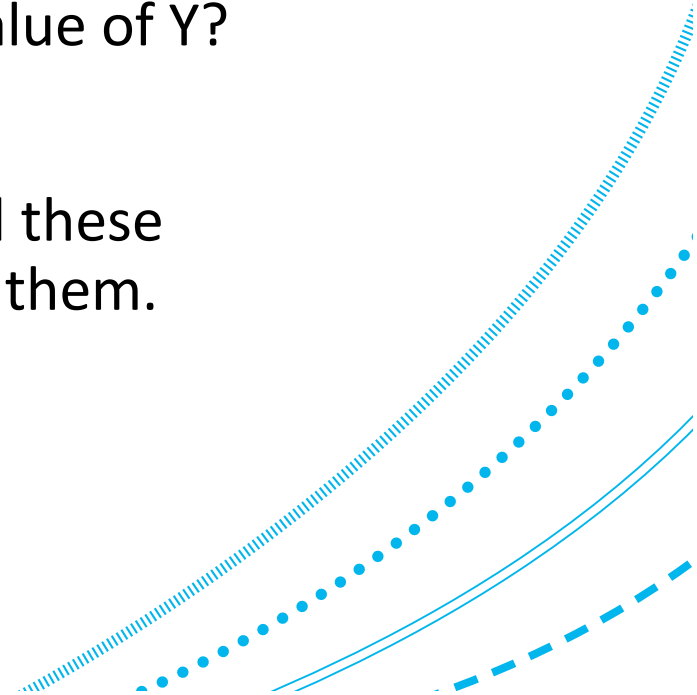


Understanding Relationships in Data

Often, there will be relationships between variables in data.

E.g. as X increases, what *typically* happens to the value of Y?

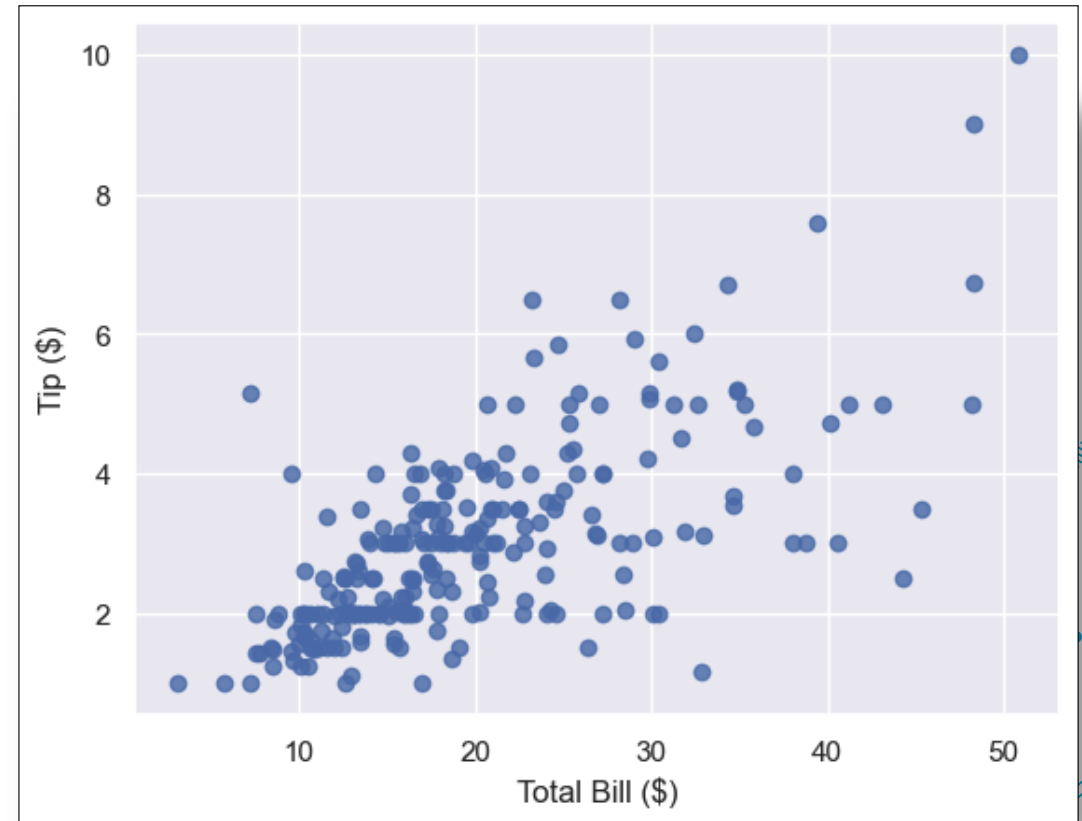
Data mining can help us to uncover and understand these relationships, and provide us with ways to visualise them.



Understanding Relationships in Data

Example: Do larger bills result in larger tips?

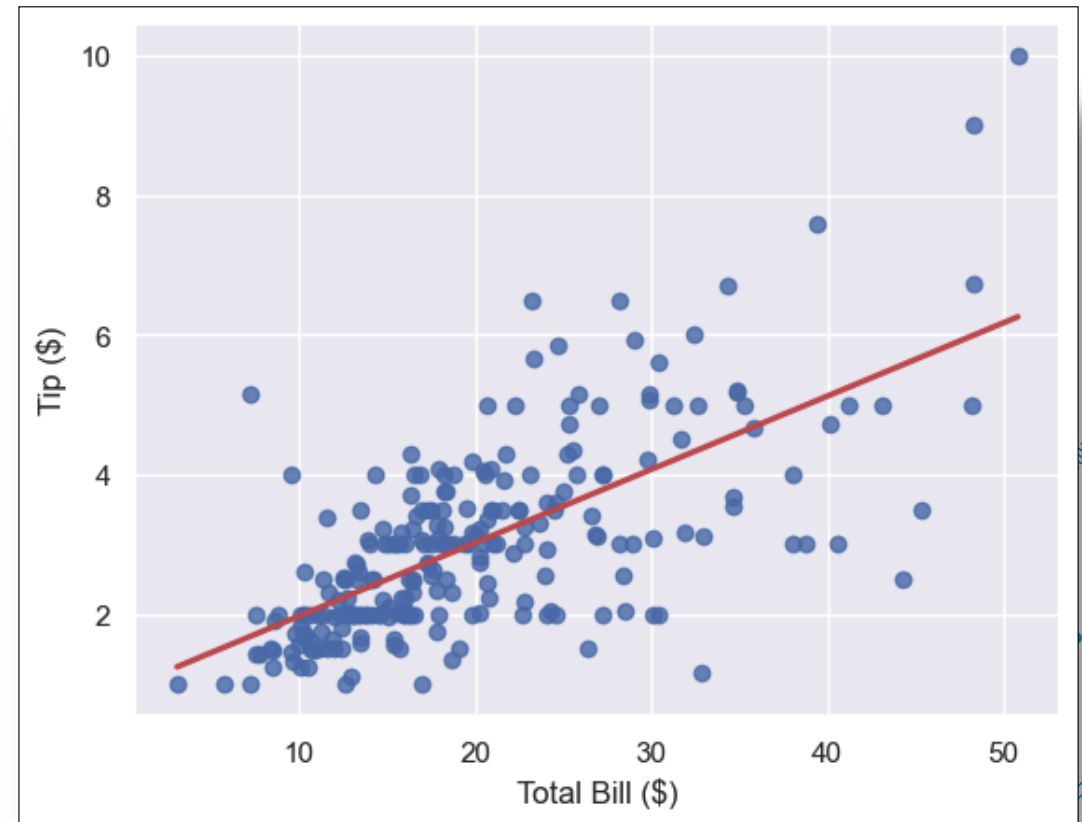
As the total bill increases, what *typically* happens to the value of tips?



Understanding Relationships in Data

Example: Do larger bills result in larger tips?

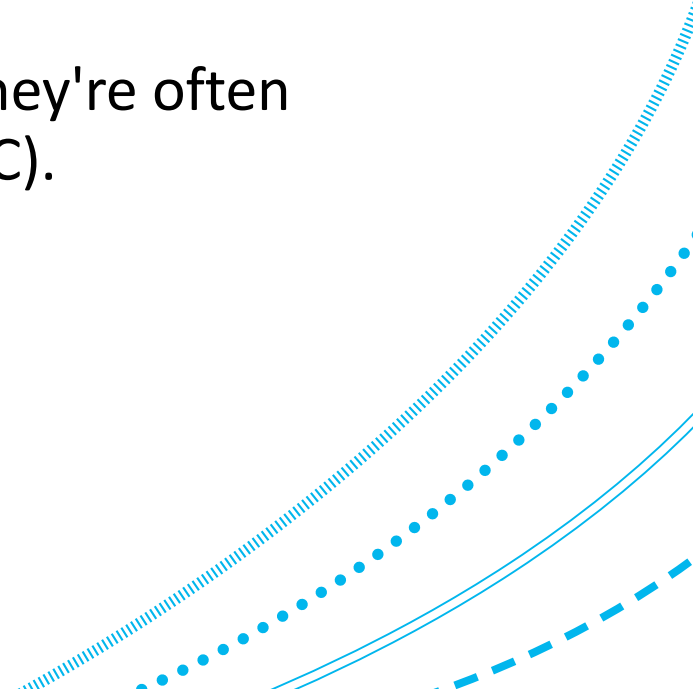
As the total bill increases, what *typically* happens to the value of tips?



Correlation

In statistics, 'correlation' is a measurement of the strength of relationship between two variables.

When people talk about a correlation coefficient, they're often referring to the Pearson Correlation Coefficient (PCC).

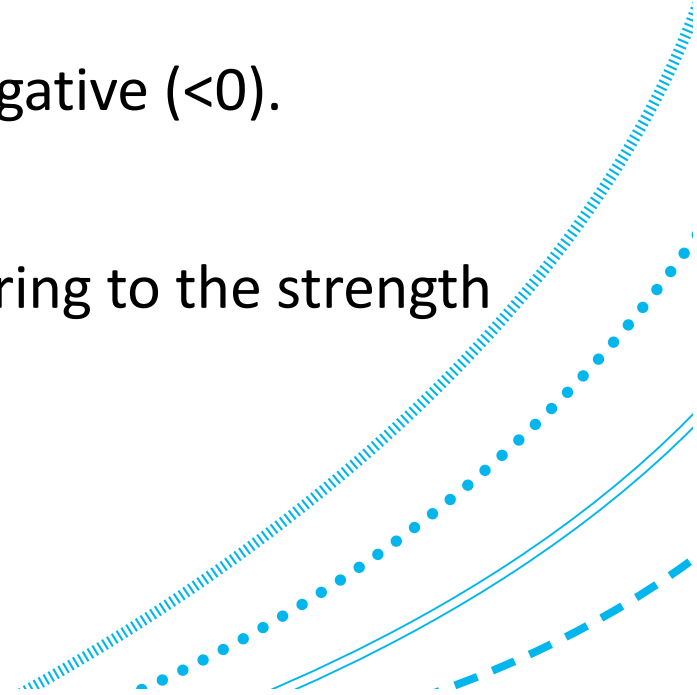


Strength and Direction

It is a numerical value between -1 and 1 , representing the strength of association.

Correlations can be described as positive (>0) or negative (<0).

They can also be described as strong or weak, referring to the strength of the correlation (either positive or negative).

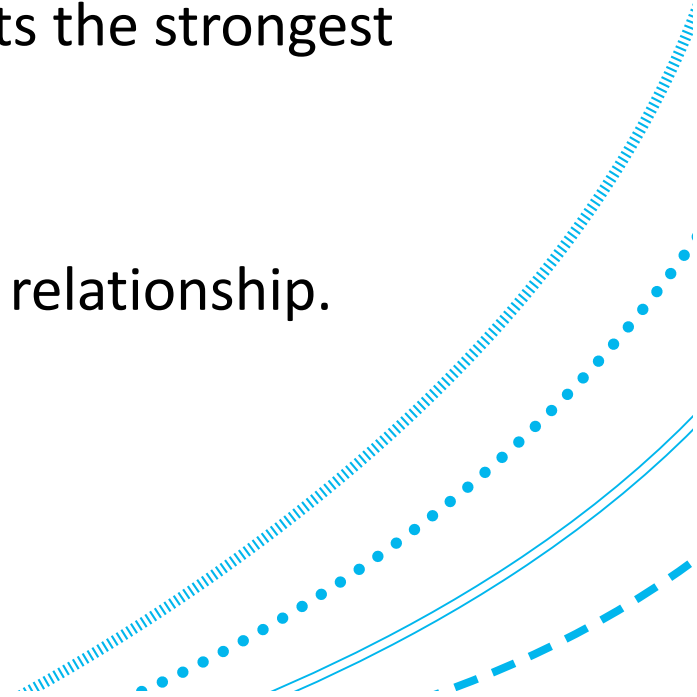


Strength and Direction

A correlation of **0** indicates no correlation between the variables.

As the correlation coefficient reaches **1**, it represents the strongest possible correlation.

A correlation coefficient of **-1** represents an inverse relationship.

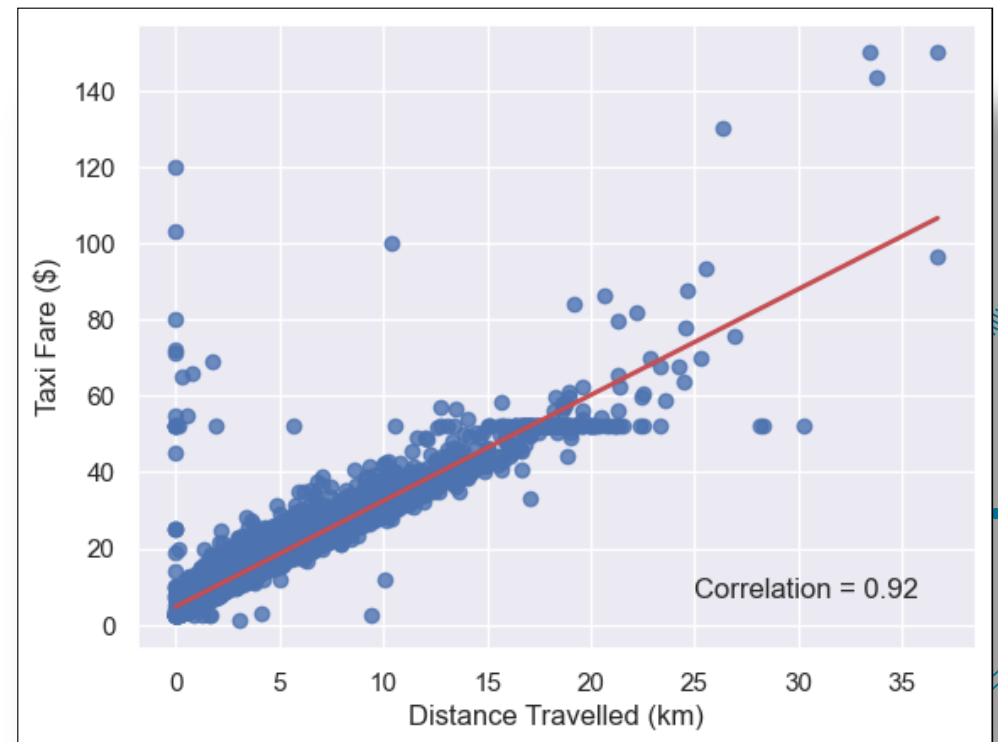


Strong Positive Correlation

Let's look at the Taxis dataset.

Is there a correlation between a taxi's fare and the distance travelled?

There is a strong positive correlation ($p=.92$) between these two variables.

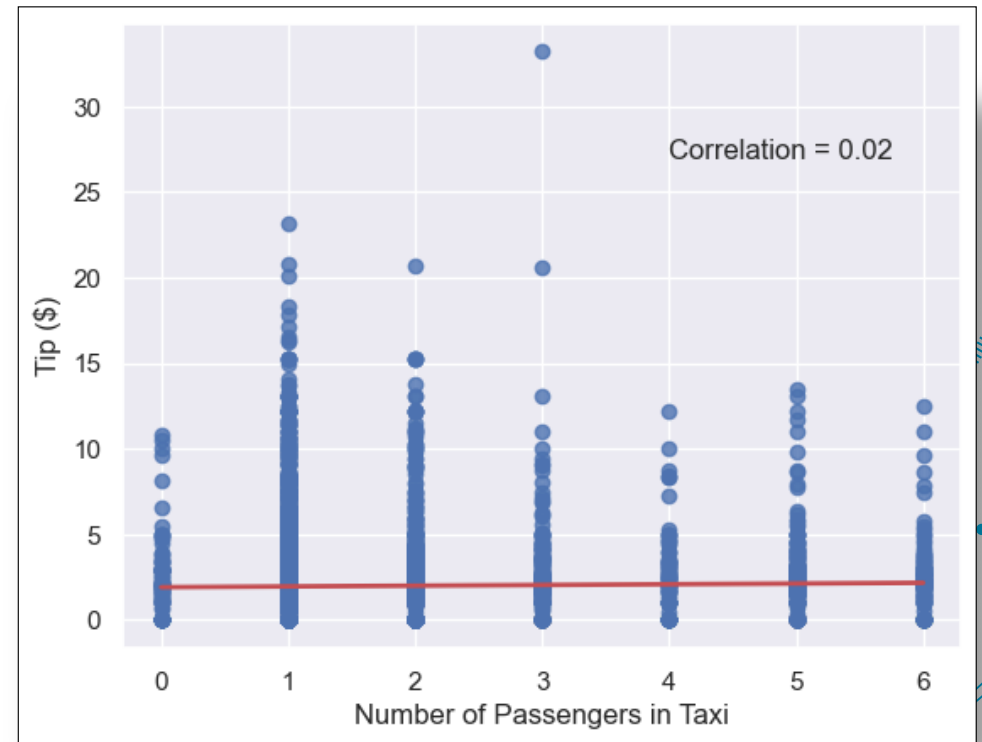


Weak Correlation

Let's look at the Taxis dataset.

Is there a correlation between the number of passengers and the amount tipped?

There is a very weak correlation ($p=.02$) between these two variables.

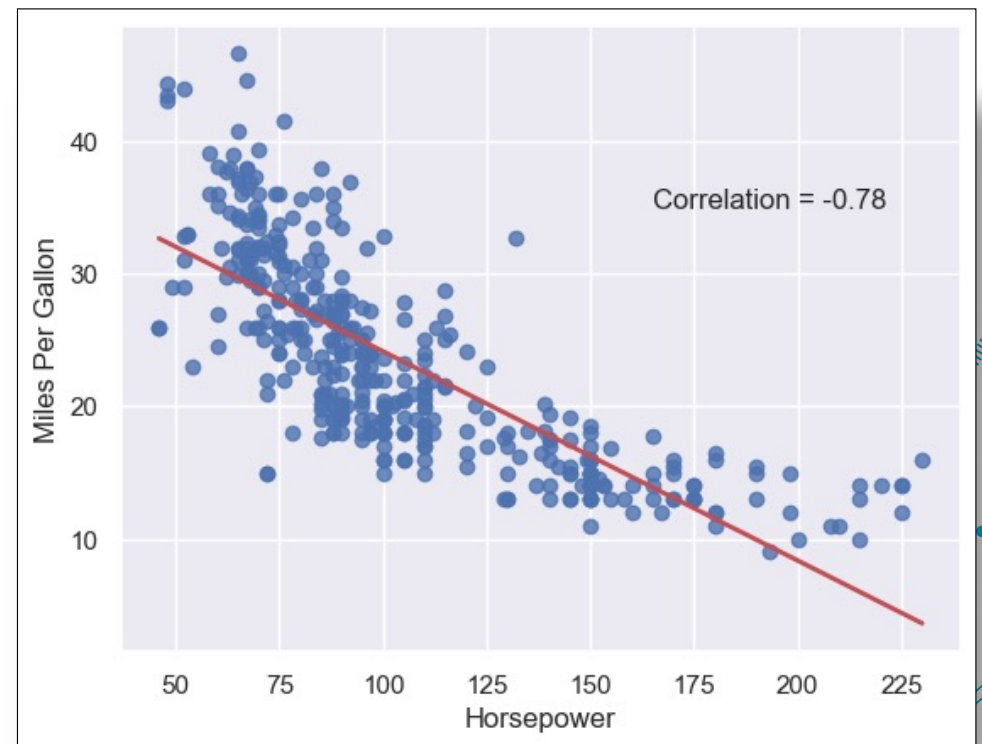


Negative Correlation

Let's look at the MPG dataset.

Is there a correlation between a car's horsepower and its miles per gallon?

There is a reasonably strong negative correlation ($p = -.78$) between these two variables.



Calculating the Pearson Correlation Coefficient of Two Variables

Note that there is a Pearson Correlation Coefficient for both **populations** and **samples**.

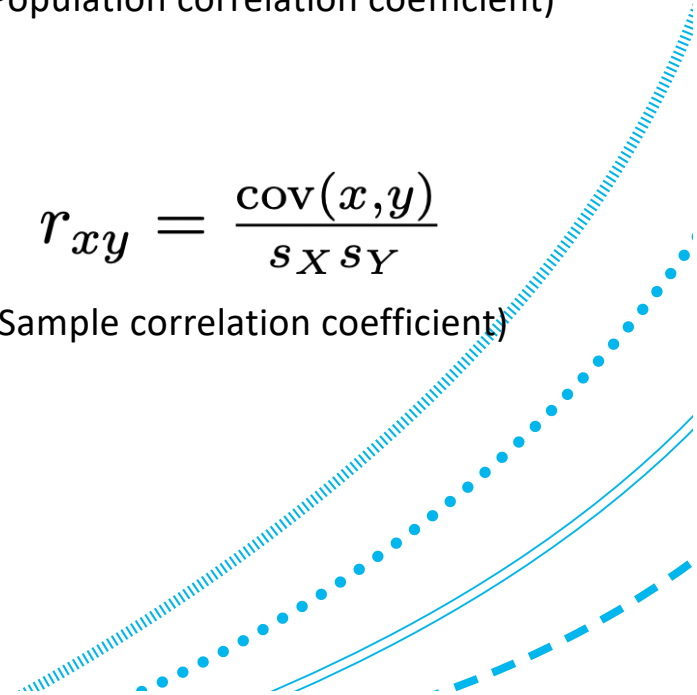
There are slight differences in how you calculate these, but for simplicity, we will just focus on samples.

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$$

(Population correlation coefficient)

$$r_{xy} = \frac{\text{cov}(x,y)}{s_X s_Y}$$

(Sample correlation coefficient)



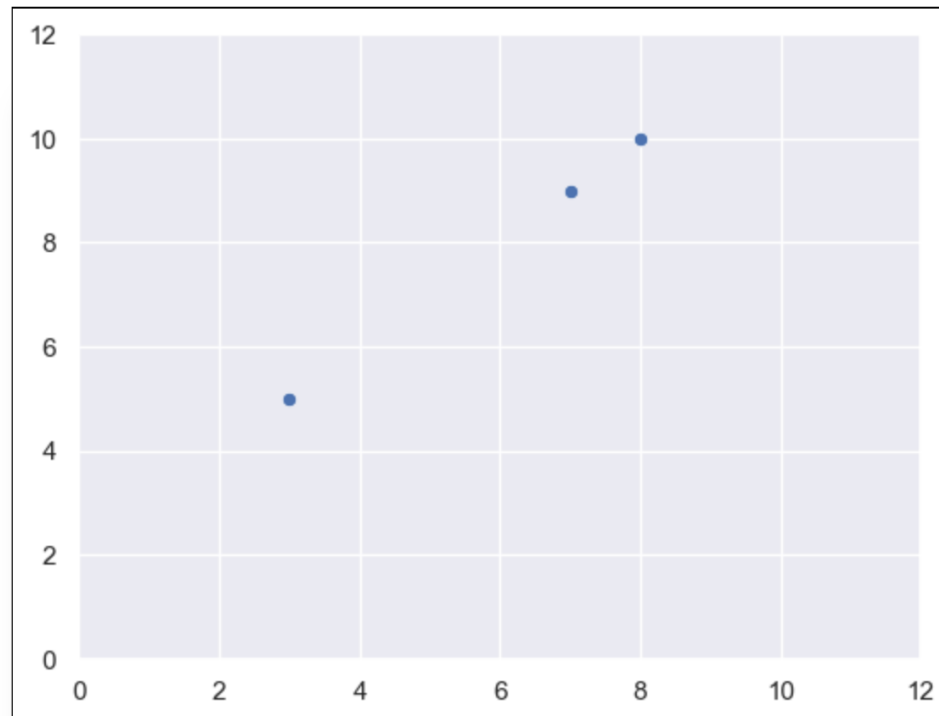
Calculating the Pearson Correlation Coefficient of Two Variables (Sample)

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{14}{\sqrt{14*14}} = \frac{14}{14} = 1$$

	x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
$\bar{x} = 6$	3	5	-3	-3	9	9	9
$\bar{y} = 8$	7	9	1	1	1	1	1
	8	10	2	2	4	4	4
Σ	18	24			14	14	14

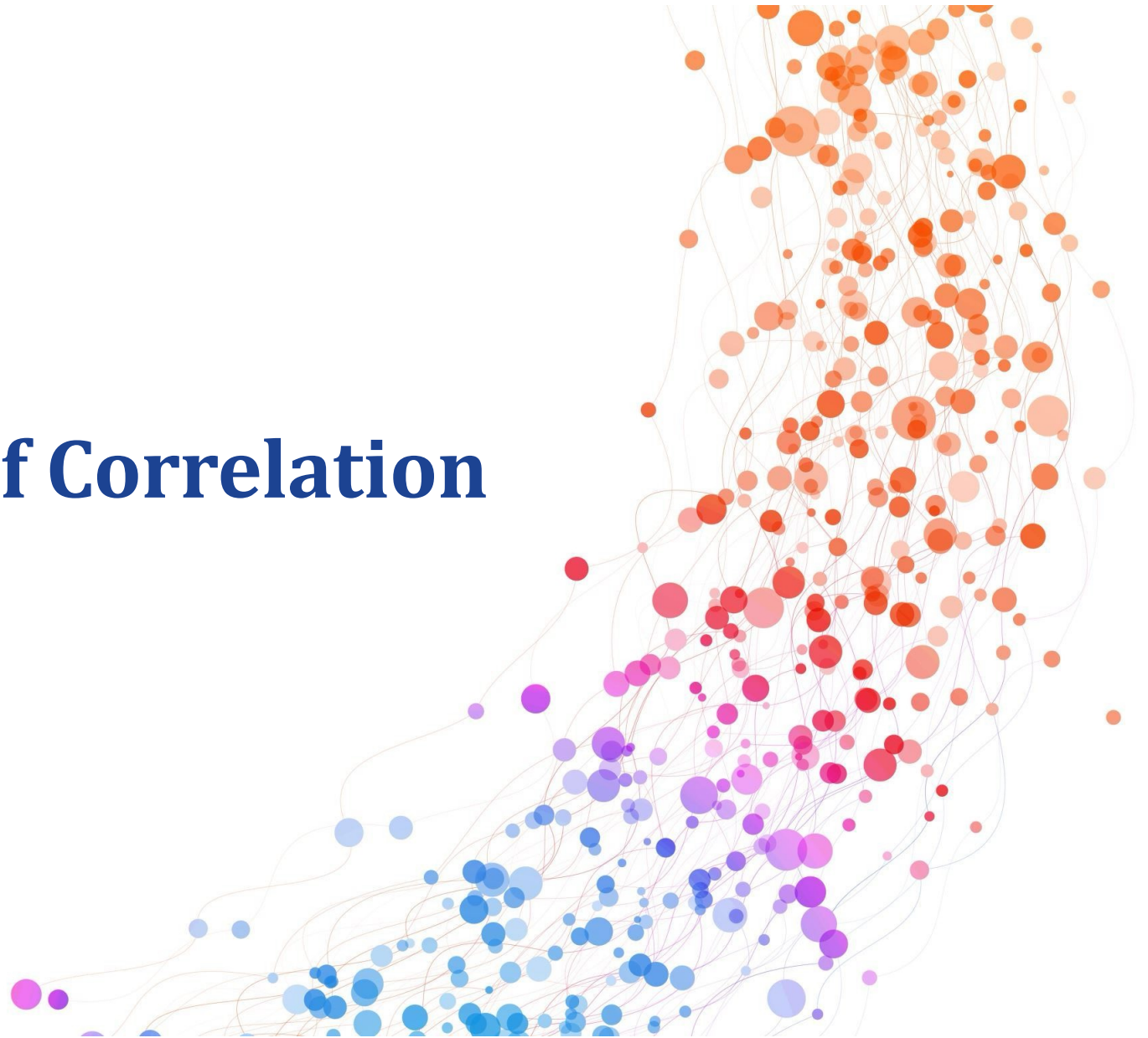
Calculating the Pearson Correlation Coefficient of Two Variables (Sample)

x	y
3	5
7	9
8	10



Limitations of Correlation

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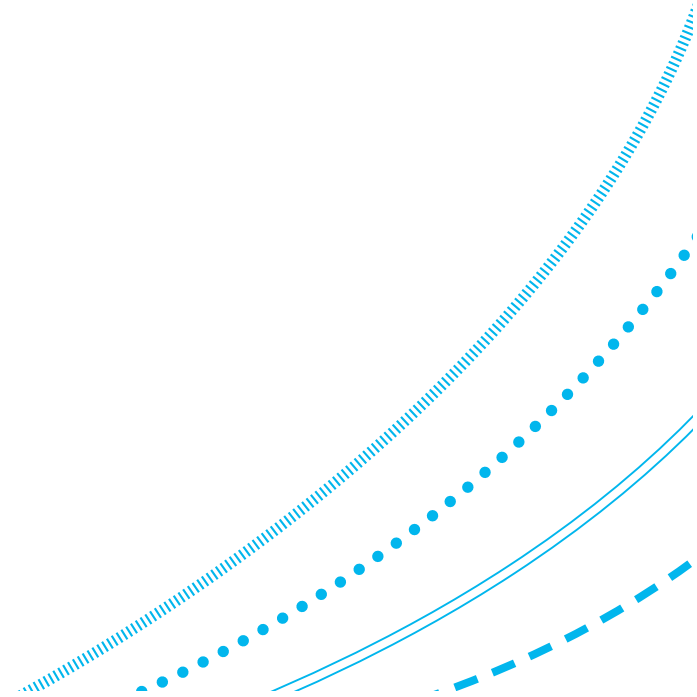


Limitations of Correlation

You may have heard the phrase '*correlation does not equal causation*'.

But what does that actually mean?

Let's look at the definitions of these...



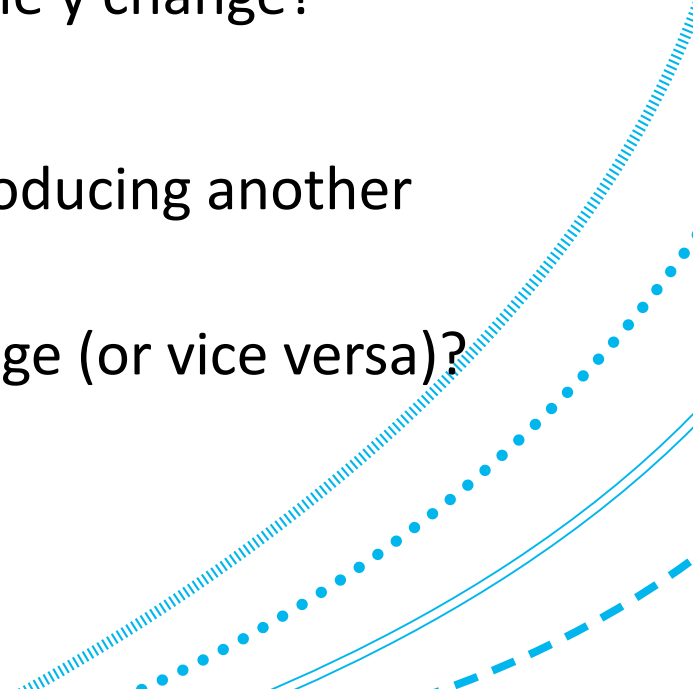
Limitations of Correlation

Correlation is a measure of the strength of an association between two variables.

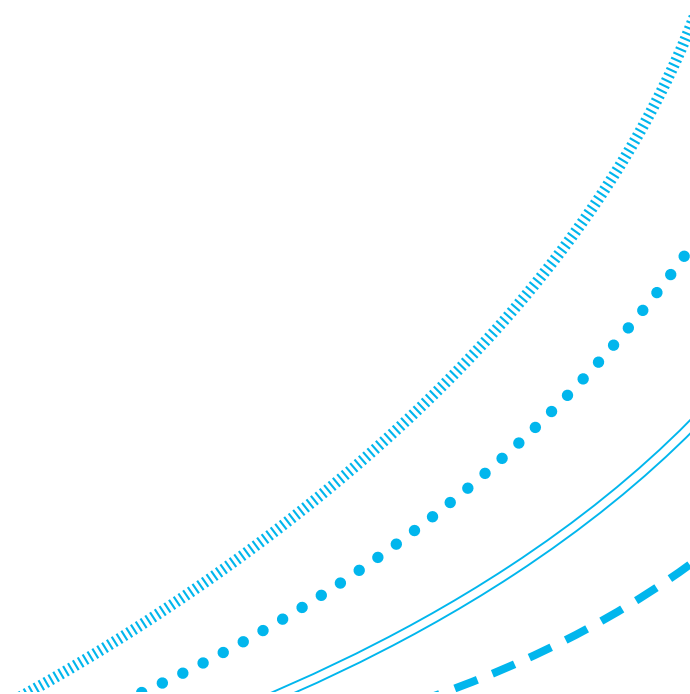
- As variable x changes, to what extent does variable y change?

Causation is the process of one event causing or producing another event.

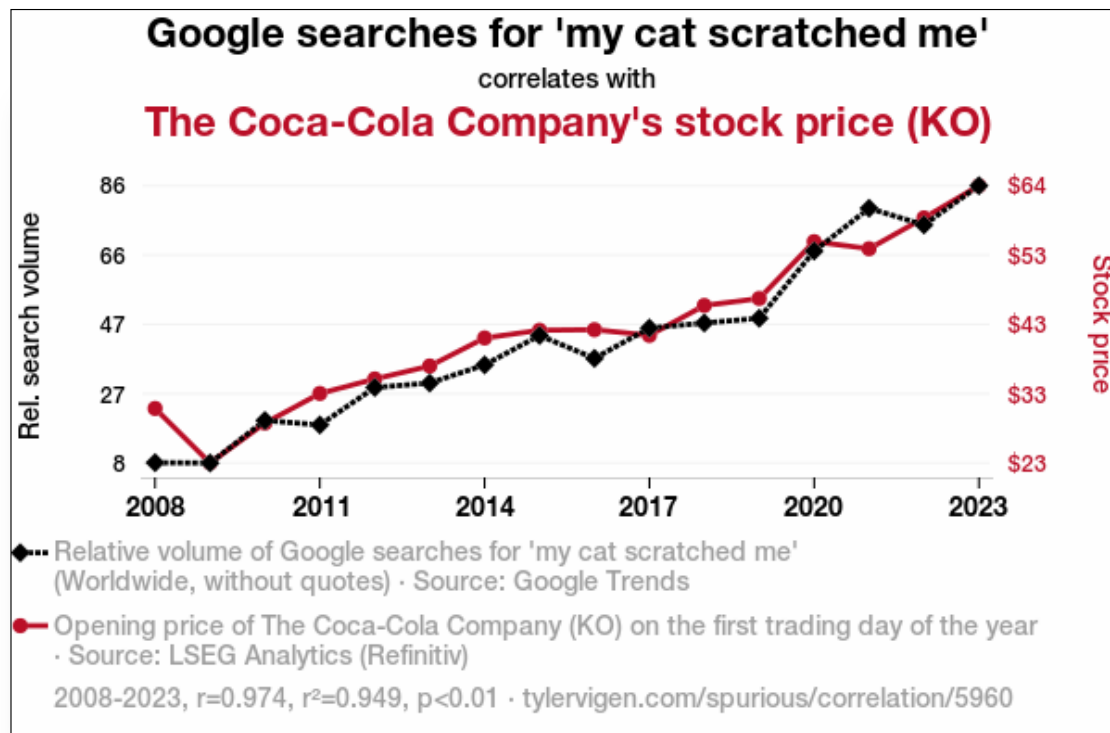
- Does changing variable x **cause** variable y to change (or vice versa)?



Limitations of Correlation



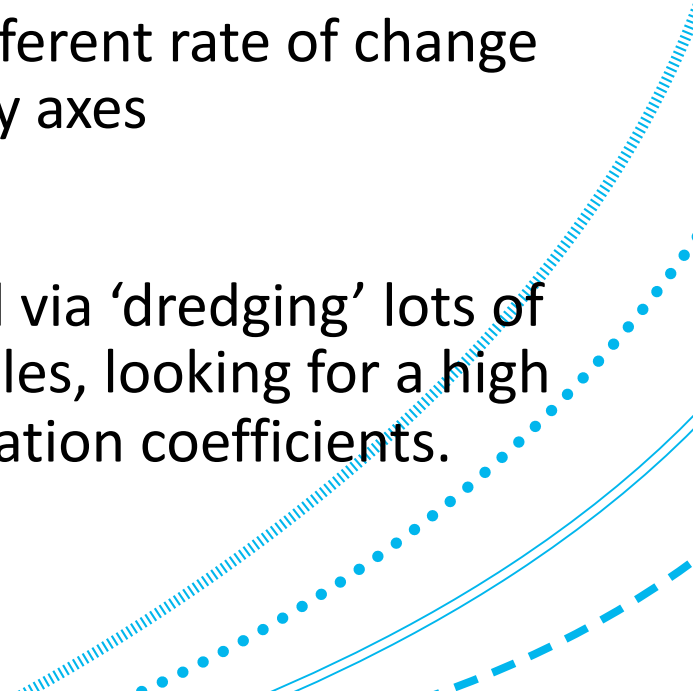
Limitations of Correlation



Note:

- Different y axes
- Y axes don't start at 0
- Different rate of change of y axes

Found via 'dredging' lots of variables, looking for a high correlation coefficients.



Limitations of Correlation

While these variables may be correlated (in some way), cat scratches probably don't cause people to buy more shares in a soft drink.

Equally, your cat *probably* doesn't care enough about your investment portfolio to get aggressive about it.

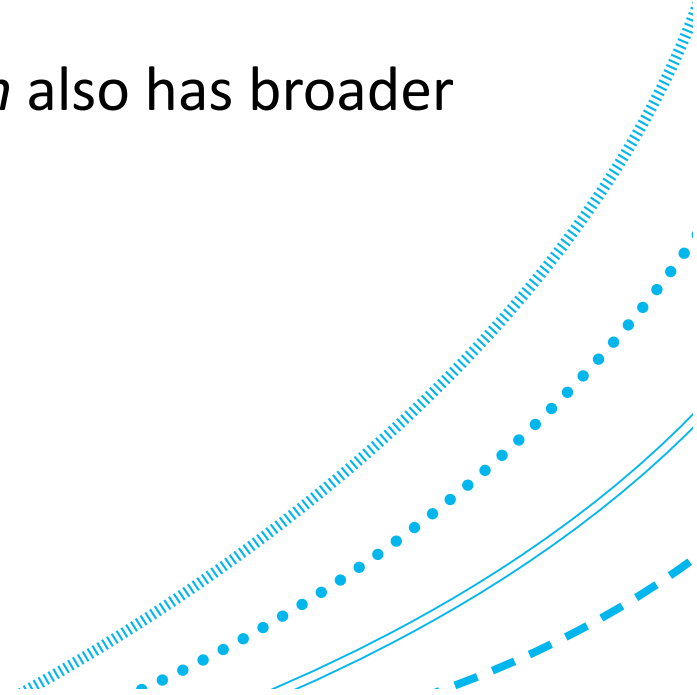


Limitations of Correlation

These spurious relationships are one example of the limits of correlation.

BUT the saying *correlation does not equal causation* also has broader implications.

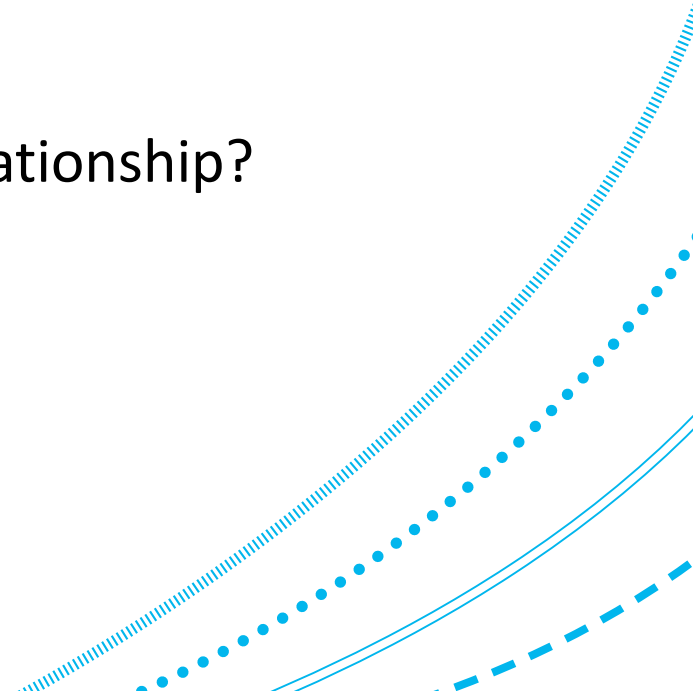
Let's take another (this time, made up) example.



Limitations of Correlation

Let's say we notice a strong positive correlation between **shark attacks** and **ice cream sales** – i.e. as the number of shark attacks increase, more ice cream is sold.

What could be the potential implications of this relationship?



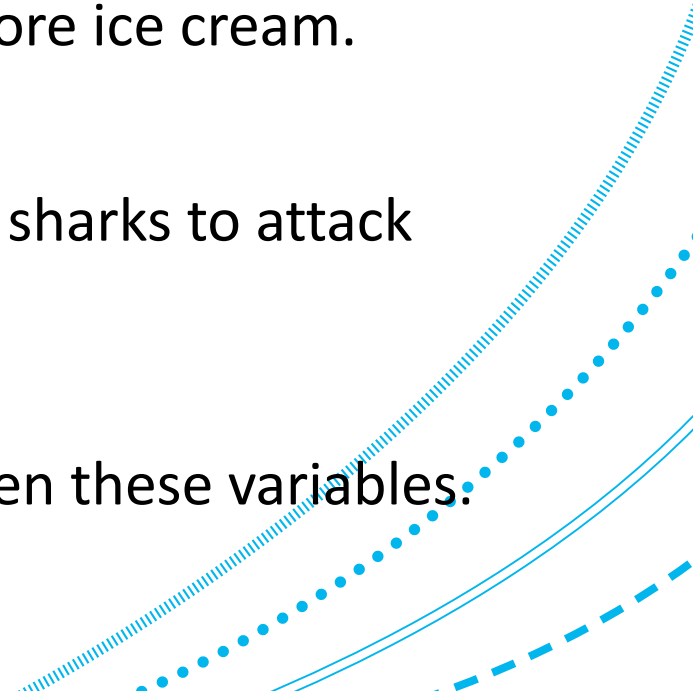
Limitations of Correlation

Possibility #1: This could be a spurious correlation

Possibility #2: Shark attacks cause people to buy more ice cream.

Possibility #3: People buying more ice cream cause sharks to attack more often

Possibility 4: There is an *indirect* connection between these variables.



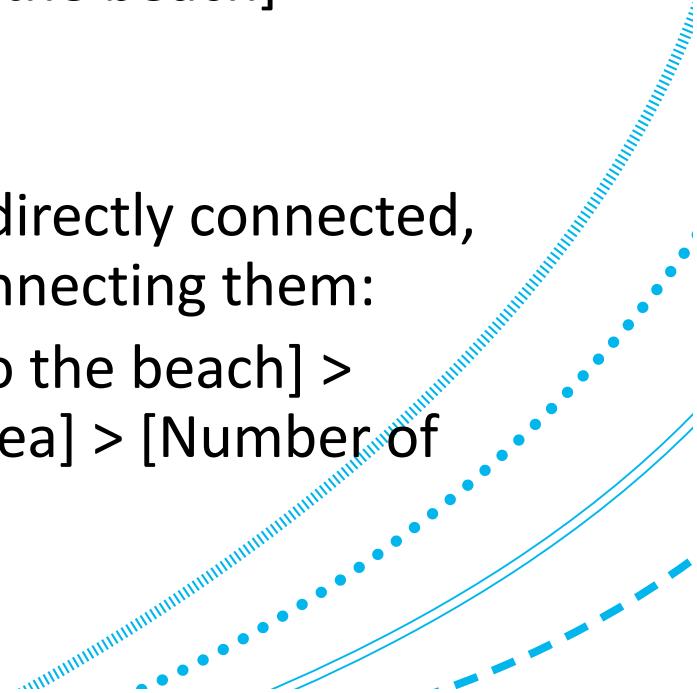
Limitations of Correlation – Direct vs Indirect

Direct correlation is where two or more variables are directly connected:

- [Temperature] > [Number of people going to the beach]

Indirect correlation is where two variables are not directly connected, but there is one or more variable that is directly connecting them:

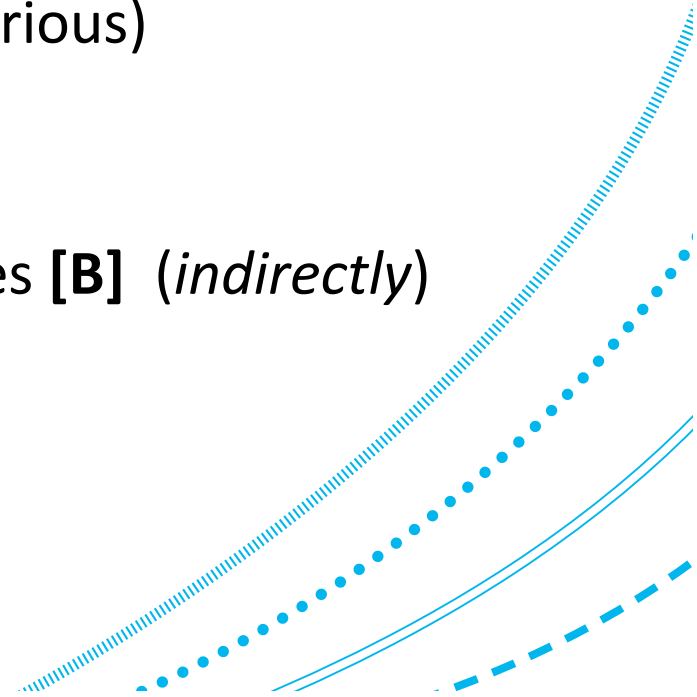
- [Temperature] > [Number of people going to the beach] > [Number of people going swimming in the sea] > [Number of shark attacks]



Limitations of Correlation

In other words, when variables **[A]** and **[B]** are correlated, it could be that:

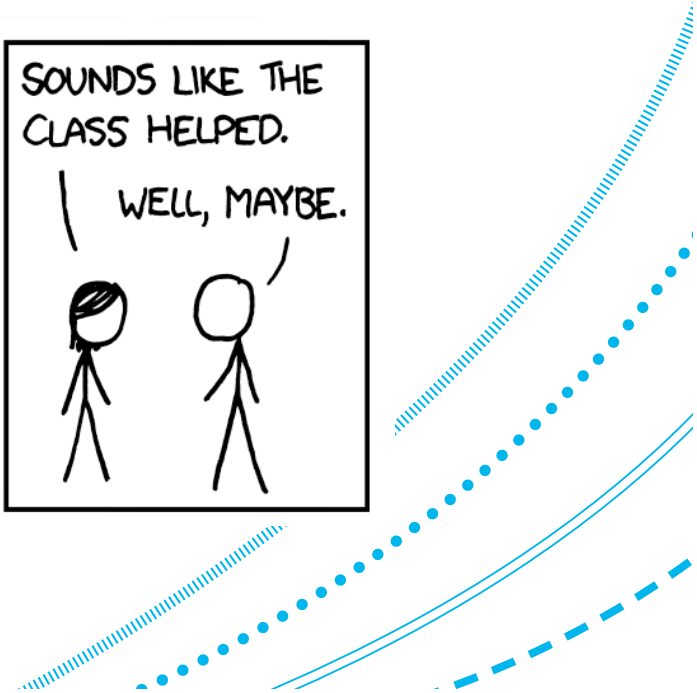
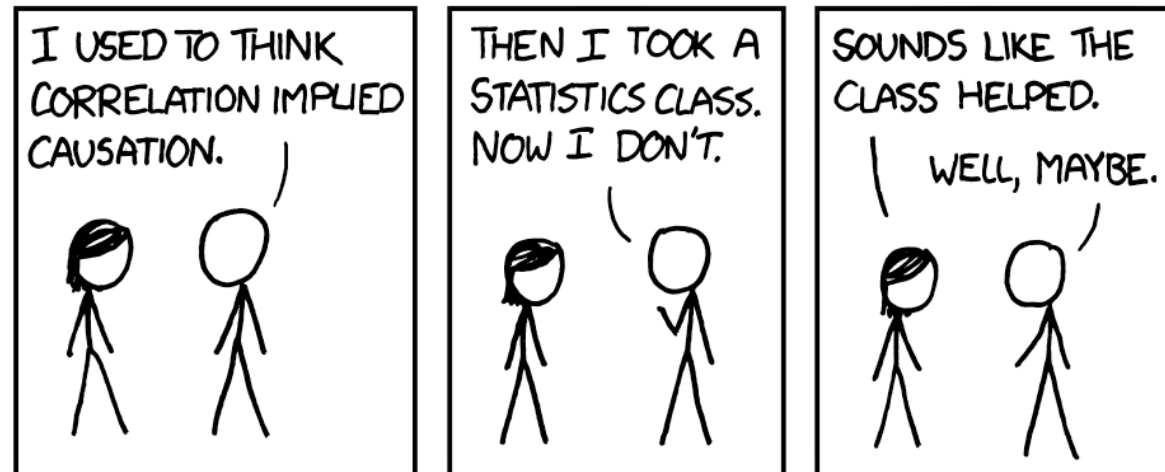
- **[A]** and **[B]** are completely unconnected (spurious)
- **[A]** causes **[B]** (*directly*)
- **[B]** causes **[A]** (*directly*)
- **[A]** causes **[D]**, which causes **[C]**, which causes **[B]** (*indirectly*)
- **[E]** causes **[A]** and **[B]** (*confounding factor*)
- Etc...



Limitations of Correlation

The only thing that the correlation tells us is that a relationship between variables *appears* to exist.

Not *why* it exists.



Limitations of Descriptive Statistics in General

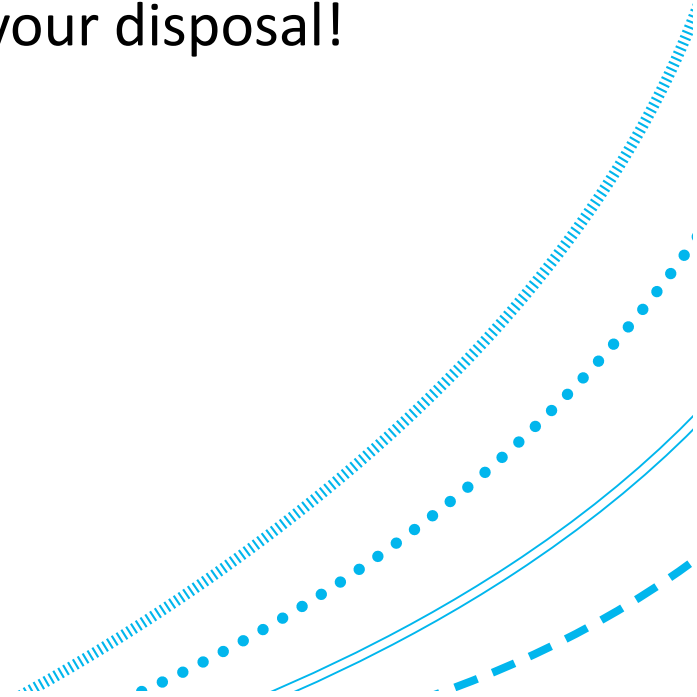
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Limitations of Descriptive Statistics

As a data scientist, descriptive statistics are invaluable.

But they are only a subset of the tools you have at your disposal!



Imagine We Have a Dataset With...

Imagine we have a dataset with:

$$\bar{x} = 9$$

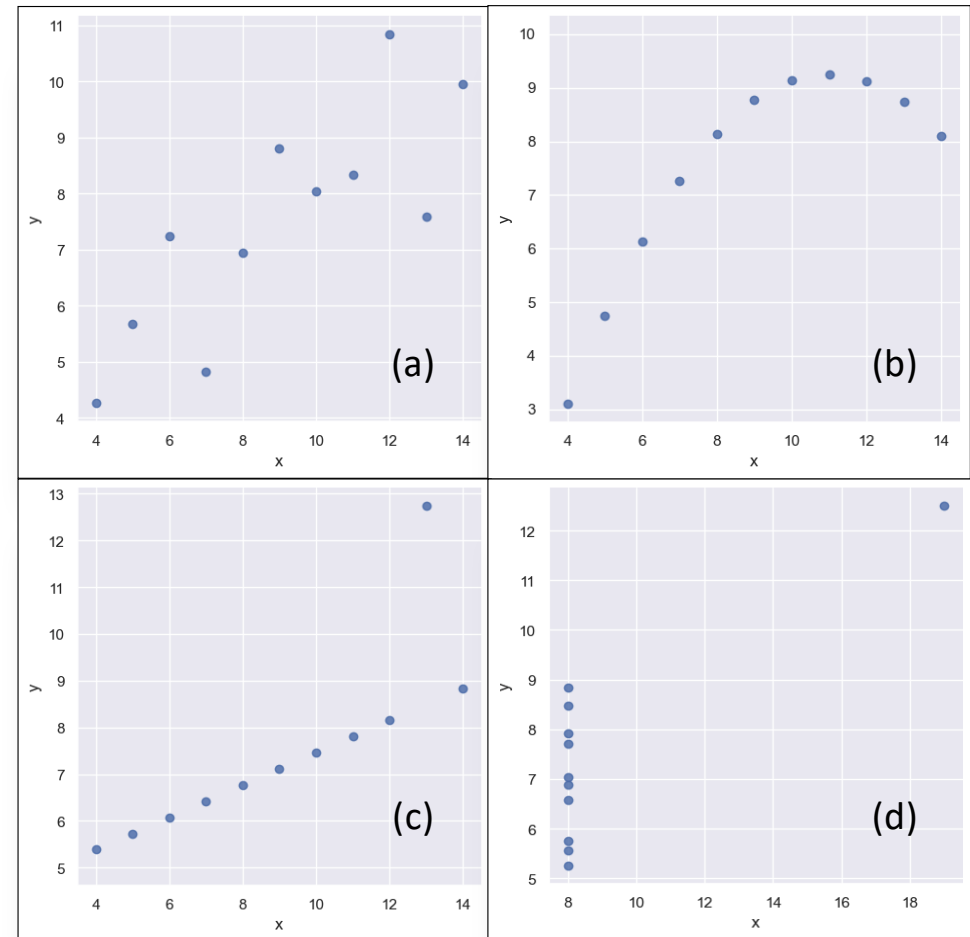
$$s_x = 3.32$$

$$\bar{y} = 7.5$$

$$s_y = 2.03$$

$$r_{xy} = 0.816$$

What does that look like?



Imagine We Have a Dataset With...

Take a minute to guess which one it is:

(a); (b); (c); or (d)

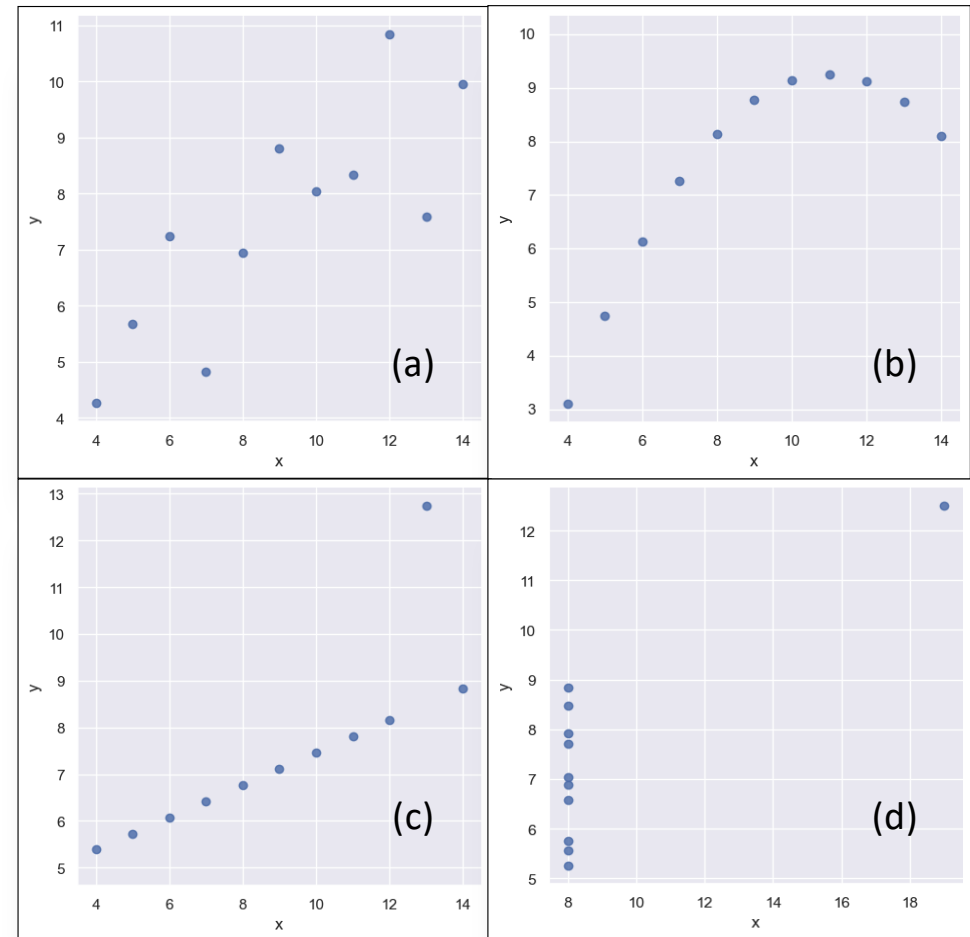
$$\bar{x} = 9$$

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$$r_{xy} = 0.816$$



Anscombe's Quartet

Answer: **All of the above!**

All four of these datasets have:

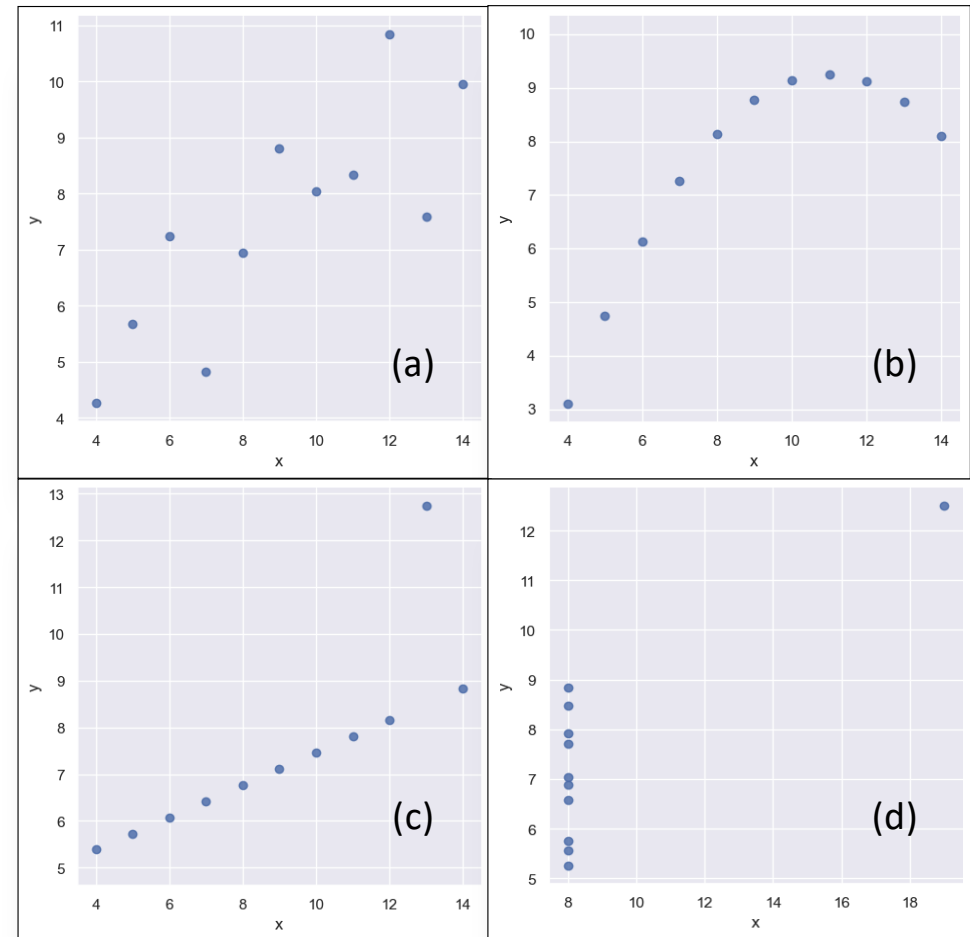
$$\bar{x} = 9$$

$$s_x = 3.32$$

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Descriptive Statistics vs Visualisations

Descriptive statistics give a 'partial picture' about your datasets.

Data visualisations can also play a vital role in helping you to gain more information and understanding about your data!

